

Project Proposal:

ISYE 521: Machine Learning in Action

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Key Questions:

How accurately can we predict apartment housing prices in Moscow with the given variables?

Since price is an indicator of preference, which apartment features do Moscow residents prioritize when searching for housing?

What features are most important to price?

Introduction to Dataset [\(link\)](#):

Our data set includes 22,700 rows and 11 features used to predict the target variable of apartment housing price in Moscow. The features include a number of categorical (#of rooms, #of floors, etc), ordinal (name of nearest metro station), binary (Region), and continuous (Area in sq. Meters) variables.

Variable Descriptions:

- **Price:** The price of the apartment in the specified currency. This is the primary target variable for prediction.
- **Apartment type:** The type of apartment, such as studio, one-bedroom, two-bedroom, etc.
- **Metro station:** The name of the nearest metro station to the apartment's location.
- **Minutes to metro:** The time in minutes required to walk from the apartment to the nearest metro station.
- **Region:** The region where the apartment is located (Moscow or Moscow Oblast).
- **Number of rooms:** The total number of rooms in the apartment, including bedrooms, living rooms, etc.
- **Area:** The total area of the apartment in square meters.
- **Living area:** The living area of the apartment in square meters, i.e., the area usable for living.
- **Kitchen area:** The area of the kitchen in square meters.
- **Floor:** The floor on which the apartment is located.
- **Number of floors:** The total number of floors in the building where the apartment is located.
- **Renovation:** The level of renovation of the apartment, such as "no renovation", "cosmetic renovation", etc.

Proposed Methodology:

- **Linear Regression:** A full model, a lasso model, and a ridge model. Regularization lambdas will be found with cross validation. Done with the scikit-learn package.
 - After running the lasso model for variable selection, putting those features into a stats. Api linear regression model for explainability.
- **Decision-Tree Regression & Random Forest Regression:** From the scikit-learn ensemble with optimization of the hyperparameters.
 - Using Grid or Random cross validation for this.
- **Neural Network Regression:** Pytorch implementation (learning later in the semester).

Prediction Target:

Apartment Housing Price in Moscow, feature name = "Price"